Executive Summary: Healthcare Operations & Financial Intelligence Dashboard

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Project: AnalystLab Africa Experience Lab – Week 3 Internship
Focus: Healthcare Operations Analysis, Patient Flow, Payer Coverage, and Strategic Business Recommendations

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1. Executive Summary & Overview

This executive summary consolidates the core findings, operational metrics, and evidence-based recommendations derived from the Week 3 Healthcare Operations & Financial Intelligence Dashboard analysis. Based on 27,891 historical patient encounters, the analysis evaluates operational volume trends, inpatient bed utilisation, encounter cost variations across service classes, and uncompensated care exposure. The insights are structured to support executive decision-making, optimize resource allocation, and strengthen financial sustainability.

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2. Key Findings

- **Encounter Volume & Class Distribution:** Outpatient and ambulatory visits represent the vast majority of operational volume, driving high patient throughput, whereas inpatient admissions while lower in volume account for disproportionately higher claim costs and resource utilization.
- **Inpatient Duration & Bed Utilization:** A significant proportion of inpatient and emergency encounters exceed 24 hours, serving as a core metric for monitoring bed turnaround times and inpatient capacity constraints.
- **Financial Exposure & Uncompensated Care:** Self-pay and uninsured encounters account for nearly **48.7%** (`Pct Zero Coverage` = 48.71%), representing a substantial financial exposure and bad-debt risk that requires structured charity care or sliding-scale eligibility verification.
- **Payer Variance:** Total claim costs vary significantly by payer, with certain commercial and public payers exhibiting wide disparities between total billed charges and recorded payer coverage.

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3. Business Recommendations (Task 5)

Recommendation 1: Implement an Advance Financial Screening & Pre-Registration Protocol for Uninsured Patients

- **Finding:** Uninsured and zero-coverage encounters account for **48.71%** of total recorded encounters (`Pct Zero Coverage`), creating severe uncompensated care exposure.
- **Proposed Action:** Deploy a front-desk pre-registration screening workflow using automated eligibility verification tools to identify self-pay patients prior to service delivery and proactively connect them with public aid or internal financial assistance programs.
- **Intended Benefit:** Reduce bad debt, lower uncompensated care write-offs, and improve hospital revenue collection predictability.
- **Limitation / Additional Info Required:** Requires cooperation from front-desk intake personnel and patient willingness to disclose financial information early; additional data on patient income levels is needed for refined means-testing.

Recommendation 2: Establish Dedicated Outpatient Fast-Track Pathways to Alleviate Emergency Department Congestion

- **Finding:** Ambulatory and outpatient classes drive high procedural volume while lower-acuity cases overlap with emergency departments, driving up average claim costs unnecessarily.
- **Proposed Action:** Reorganize facility scheduling and triage protocols to divert low-acuity walk-in encounters into dedicated ambulatory fast-track lanes, separating them from acute emergency trauma cases.
- **Intended Benefit:** Decrease emergency department wait times, optimize physician time allocation, and reduce overall cost per encounter for minor treatments.
- **Limitation / Additional Info Required:** Depends on staffing availability during peak hours and physical facility layout constraints.

Recommendation 3: Conduct a Payer-Specific Contract and Reimbursement Audit

- **Finding:** Significant cost variations exist across commercial and public payers relative to actual payer coverage amounts, indicating potential under-reimbursement gaps.
- **Proposed Action:** Initiate a formal audit of third-party payer contracts comparing contracted rate schedules against actual settled claim reimbursements to identify denied claims and systemic under-payments.
- **Intended Benefit:** Enhance revenue recovery, plug leakage from under-compensating payer agreements, and strengthen future contract negotiation leverage.

- **Limitation / Additional Info Required:** Requires historical remittance advice data and detailed denial reason codes which were partially aggregated in the primary dataset.

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4. Deliverables Checklist & Verification

1. **Power BI Working Dashboard:** Fully structured widescreen (1280x720) report containing top-level KPI cards, encounter trend line charts, encounter class breakdown columns, and payer financial exposure bar charts linked to the centralized `_Measures` table.
2. **Data-Quality Assessment:** Documented missing timestamps, handling of zero-coverage values, and relational integrity across `encounters`, `patients`, `procedures`, and `payers`.
3. **KPI Definition Sheet:** Standardized definitions for `Total Encounters`, `Pct Encounters Over 24 Hours`, `Total Claim Cost`, `Avg Claim Cost`, `Total Payer Coverage`, and `Pct Zero Coverage`.
4. **Dashboard Layout & Screenshots:** Verified executive layout comprising header notes, left-hand slicer sidebar, and analytical visual grids.
5. **Validation Evidence:** Cross-reconciled total row counts (27,891 encounters) and verified dynamic filter interactions across all visual elements.
6. **Executive Summary:** This comprehensive document satisfying Task 5 requirements.